KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

YEAR 10 TRANSITION

EXTENSION BOOKLET

FOR FAST FINISHERS · CHALLENGE QUESTIONS

CHAPTER 1

YEAR 10 MATHEMATICS · CHAPTER 1 · EXTENSION: FAST FINISHERS

Financial mathematics

One page of harder questions for each exercise in Chapter 1. Finish the exercise in your Chapter 1 booklet first, then come here. Worked answers are at the back.

EXERCISE	EXTENSION TOPIC	PAGE	DONE
1A	Review of percentages	2	☐
1B	Applying percentages	3	☐
1C	Income	4	☐
1D	Income tax	5	☐
1E	Budgeting	6	☐
1F	Simple interest	7	☐
1G	Compound interest	8	☐
1H	Investments and loans	9	☐
1I	Comparing interest using technology	10	☐

EVERY CARD HAS A TYPE

Understanding
Reasoning
Problem solving

MIXED IN

Review 1A questions from earlier exercises

NO DIAGRAM?

✎ **Draw a diagram** first, in the dotted space

CHALLENGE

★☆☆ to ★★★

The last card on each page is the challenge.

NAME _____ CLASS _____ TEACHER _____

Artwork: a spiral from the tail of a seahorse in Seahorse Valley, part of the Mandelbrot set, coloured gold.

Review of percentages

FAST FINISHERS

EXTENSION Finished exercise 1A? Try these. Show your working in each box. Round percentages to **one decimal place** where needed.

Understanding · Reasoning · Problem solving

1 Understanding ★☆☆

Copy and complete the table.

Fraction	Decimal	Percentage
3/8		
	0.45	
		2.5%
1¼		

2 Understanding ★☆☆

These are all close in size.

a Write each as a decimal: 0.63, 5/8, 61%, 2/3

b Write them in order, smallest to largest.

c Which is larger, 7/20 or 34%? Show how you know.

b _____ c _____

3 Reasoning ★★☆

Find the error. Sam worked out 12% of $250 like this:

12% of $250 = 250 ÷ 12 = $20.83

a Explain Sam's mistake.

b Find the correct answer.

c Show a quick way to check: find 10% and 2% of $250 and add.

b _____ c _____

4 Problem solving ★★☆

In a class of 28 students, 9 walk to school, 12 catch the bus and the rest are driven.

a What percentage walk?

b What percentage are driven?

c Do the three percentages add to 100%? Explain.

a _____ b _____

5 Reasoning ★★☆

Mia scored 42 out of 56 in Test A and 33 out of 42 in Test B.

a Write each score as a percentage.

b In which test did she do better? Explain why comparing 42 and 33 is not enough.

A _____ B _____ better _____

6 Problem solving ★★★

Work backwards. 15% of a number is 36.

a Find 1% of the number.

b Find the number.

c Find 40% of the number.

d Explain a different way to get c from 36.

a _____ b _____ c _____

Applying percentages

FAST FINISHERS

EXTENSION Finished exercise 1B? Try these. Show your working in each box. Round money to the **nearest cent**. GST is 10%.

Understanding | Reasoning | Problem solving

1 Understanding ★☆☆

a Increase $80 by 15%.

b Decrease $64 by 25%.

c Write the single number you multiply by to increase a price by 7%.

d Write the multiplier to decrease a price by 12%.

a _____ b _____ c _____ d _____

2 Understanding | Review 1A ★☆☆

a Write 0.085 as a percentage.

b Write 0.085 as a fraction in simplest form.

c Write 3/40 as a percentage.

a _____ b _____ c _____

3 Reasoning ★★☆

The same jacket is sold in two shops.

	Price	Sale
Shop A	$120	25% off
Shop B	$108	$20 off

a Find the sale price at each shop.

b Which shop is cheaper, and by how much?

A _____ B _____ b _____

4 Problem solving ★★☆

Work backwards. A TV is on sale for $630 after a 30% discount.

a What percentage of the original price is $630?

b Find the original price.

c Ali says, "Just add 30% of $630 back on." Show that this gives the wrong answer.

a _____ b _____ c _____

5 Understanding ★★☆

a A meal costs $48 before GST. Find the price including GST.

b A phone costs $319 **including** GST. Find the price before GST.

c How much GST is in the $319?

a _____ b _____ c _____

6 Reasoning ★★★

A price of $200 is increased by 10%. The new price is then decreased by 10%.

a Find the final price.

b Explain why the price does not go back to $200.

c What single percentage change gives the same final price?

a _____ c _____

EXTENSION Finished exercise 1C? Try these. Show your working in each box. Use 52 weeks, 26 fortnights or 12 months in a year. Understanding | Reasoning | Problem solving

1 Understanding ★☆☆

Ava earns a salary of $67 600 per year.

a Find her weekly pay.

b Find her fortnightly pay.

c Find her monthly pay.

a _____ b _____ c _____

2 Understanding ★☆☆

Noah is paid $24.80 per hour.

a Find his pay for a 38-hour week.

b Find his time-and-a-half rate.

c Find his double-time rate.

a _____ b _____ c _____

3 Problem solving ★★☆

Jack is paid $28 per hour. One week he works 38 normal hours, 4 hours at time and a half, and 3 hours on Sunday at double time.

a Find his normal pay.

b Find his overtime pay.

c Find his total pay for the week.

a _____ b _____ c _____

4 Reasoning ★★☆

Job A pays $1150 per week. **Job B** pays a retainer of $600 per week plus 5% commission on sales.

a Find the weekly pay for Job B with sales of $9000.

b What sales are needed for Job B to pay the same as Job A?

c Which job would you choose? Give a reason.

a _____ b _____

5 Understanding Review 1B ★★☆

Zoe earns $26 per hour. She gets a 4% pay rise.

a Find her new hourly rate.

b Find her new pay for a 35-hour week.

c How much more does she earn each week?

a _____ b _____ c _____

6 Problem solving ★★★

Work backwards. Mel earned $1092 in one week. She worked 36 normal hours and 4 hours at time and a half.

a Explain why 4 hours at time and a half is worth the same as 6 normal hours.

b Find Mel's hourly rate.

c Check your answer by working out her pay.

b _____ c _____

Income tax

FAST FINISHERS

EXTENSION Finished exercise 1D? Try these. Show your working in each box. Use the tax table below.

Understanding · Reasoning · Problem solving

TAXABLE INCOME	TAX ON THIS INCOME (2024–25 resident rates)
$0 – $18 200	Nil
$18 201 – $45 000	16c for each $1 over $18 200
$45 001 – $135 000	$4288 plus 30c for each $1 over $45 000
$135 001 – $190 000	$31 288 plus 37c for each $1 over $135 000
$190 001 and over	$51 638 plus 45c for each $1 over $190 000

Medicare levy = 2% of taxable income. **Taxable income** = gross income – deductions.

1 Understanding ★☆☆

Liam earns $58 000 a year. He claims $1400 of deductions for his uniform and union fees.

a Find his taxable income.

b Find the tax he pays.

a _____ b _____

2 Understanding ★☆☆

a Find the tax on a taxable income of $30 000.

b Find the tax on a taxable income of $16 500.

c Explain your answer to b.

a _____ b _____

3 Reasoning ★★☆

Find the error. Zara says:

"I earn $50 000, so I pay 30% tax. 0.3 × 50 000 = $15 000."

a Explain Zara's mistake.

b Find the correct tax.

b _____

4 Problem solving ★★☆

Ben's taxable income is $72 000.

a Find his income tax.

b Find his Medicare levy.

c Find the total he pays.

d What percentage of his income is this?

a _____ b _____ c _____ d _____

5 Problem solving ★★☆

Ruby's employer takes $420 tax (PAYG) out of each fortnightly pay. Her taxable income for the year is $62 000.

a How much tax was taken out over the year?

b Find her tax plus Medicare levy.

c Will she get a refund or owe money? How much?

a _____ b _____ c _____

6 Reasoning ★★★

Work backwards. Tom paid $6088 income tax (not counting Medicare).

a Explain how you know his taxable income is more than $45 000.

b Find his taxable income.

b _____

Budgeting

FAST FINISHERS

EXTENSION Finished exercise 1E? Try these. Show your working in each box. Use 52 weeks, 26 fortnights or 12 months in a year.

Understanding · Reasoning · Problem solving

1 Understanding ★☆☆

Convert each amount.

a $45 per week to per year
b $1200 per quarter to per year
c $1200 per quarter to per month
d $180 per fortnight to per week

a _____ b _____ c _____ d _____

2 Understanding ★☆☆

Kai's weekly net pay is $1150.

Expense	Per week
Rent	$420
Food	$160
Transport	$65
Phone	$15
Fun	$90

a Find his total weekly expenses.
b How much is left over each week?
c What percentage of his pay goes on rent?

a _____ b _____ c _____

3 Reasoning Review 1C ★★☆

Ella earns $23.50 per hour and works 30 hours a week. She saves 15% of her pay.

a Find her weekly pay.
b How much does she save each week?
c How many weeks to save $1000?

a _____ b _____ c _____

4 Problem solving ★★☆

Milan's car costs:

Registration	$780 per year
Insurance	$96 per month
Servicing	$450 every 6 months
Fuel	$55 per week

a Find the total cost for one year.
b Find the average cost per week.

a _____ b _____

5 Reasoning ★★☆

Ali's fortnightly net pay is $1400. He budgets 30% for rent, 20% for food, 15% for transport, 10% for savings and the rest for other things.

a What percentage is for other things?
b How much does he save each fortnight?
c How many fortnights to save $2100 for a holiday?

a _____ b _____ c _____

6 Problem solving ★★★

Sophie wants a $1350 laptop in 12 weeks. She already has $270. She earns $310 a week and spends $235.

a How much does she need to save each week to reach her goal?
b Will she reach her goal in 12 weeks? Show working.
c If not, how many weeks will it take?

a _____ b _____ c _____

Simple interest

FAST FINISHERS

EXTENSION Finished exercise 1F? Try these. Show your working in each box. Use $I = PRN$. Round money to the **nearest cent**.

Understanding | Reasoning | Problem solving

1 Understanding ★☆☆

$3000 is invested at 4% p.a. simple interest for 5 years.

a Find the interest.

b Find the total value of the investment.

a _____ b _____

2 Understanding ★☆☆

$2500 is invested at 6% p.a. simple interest.

a Find the interest for 18 months.

b Find the interest for 9 months.

c Explain why b is half of a.

a _____ b _____

3 Reasoning ★★☆

Find the error. Jo finds the simple interest on $800 at 5% p.a. for 3 years:

$I = 800 \times 5 \times 3 = \$12\,000$

a Explain why the answer must be wrong.

b Find the error and the correct interest.

b _____

4 Problem solving ★★☆

Work backwards. $4000 is invested for 3 years and earns $540 simple interest.

a Find the interest earned in one year.

b Find the interest rate.

a _____ b _____

5 Problem solving ★★☆

Work backwards. How long will it take $2000 to earn $350 simple interest at 3.5% p.a.?

a Find the interest for one year.

b Find the number of years.

a _____ b _____

6 Reasoning Review 1E ★★★

Max borrows $6000 for a car at 8% p.a. simple interest for 3 years. He repays it in equal monthly amounts.

a Find the interest.

b Find the total amount he repays.

c Find the monthly repayment.

d Max's budget allows $230 a month for the car. Can he afford it?

a _____ b _____ c _____ d _____

Compound interest

FAST FINISHERS

EXTENSION Finished exercise 1G? Try these. Show your working in each box. Use $A = P(1 + R)^n$. Round money to the **nearest cent**.

Understanding · Reasoning · Problem solving

1 Understanding ★☆☆

$1000 is invested at 5% p.a., compounded annually. Copy and complete the table.

Year	Start	Interest	End
1	$1000		
2			
3			

2 Understanding ★☆☆

$5000 is invested at 6% p.a., compounded annually, for 4 years.

a Find the final amount A.

b Find the interest earned.

a _____ b _____

3 Reasoning Review 1F ★★☆

$4000 is invested for 3 years at 5% p.a.

a Find the simple interest.

b Find the compound interest (compounded annually).

c Explain why compound interest gives more.

a _____ b _____

4 Understanding ★★☆

$2000 is invested at 6% p.a., **compounded monthly**, for 2 years.

a Find the rate per month, R, as a decimal.

b Find the number of months, n.

c Find the final amount.

a _____ b _____ c _____

5 Reasoning ★★☆

Find the error. To find $3000 at 4% p.a. compounded annually for 2 years, Kim writes:

$A = 3000(1 + 4)^2 = \$75\,000$

a Explain Kim's mistake.

b Find the correct amount.

b _____

6 Problem solving ★★★

$5000 is invested at 8% p.a., compounded annually.

a Find its value after 5 years.

b Try different numbers of years to find when it first becomes more than $10 000.

c Would it take twice as long to reach $20 000? Explain.

a _____ b _____

Investments and loans

FAST FINISHERS

EXTENSION Finished exercise 1H? Try these. Show your working in each box. Round money to the **nearest cent**.

Understanding | Reasoning | Problem solving

1 Understanding ★☆☆

Chloe borrows $15 000 for a car. She repays $380 per month for 4 years.

a Find the total amount she repays.

b Find the interest she pays.

a _____ b _____

2 Understanding ★☆☆

A fridge costs $1800. It can be bought with a 10% deposit, then $85 per month for 2 years.

a Find the deposit.

b Find the total amount paid.

c How much more is this than the cash price?

a _____ b _____ c _____

3 Reasoning Review 1F ★★☆

Look at Chloe's loan in question 1.

a Find the interest she pays each year.

b Find the simple interest rate for the loan.

a _____ b _____

4 Problem solving ★★☆

Dev borrows $1000. Interest of 1% is added each month, then he repays $200. Copy and complete the table.

Month	Start	Interest	Repay	Owing
1	$1000.00		$200	
2			$200	
3			$200	

5 Reasoning ★★☆

Lena has $10 000 to invest for 3 years.
Option A: 4.2% p.a. simple interest
Option B: 4% p.a. compounded annually

a Find the interest for each option.

b Which option is better, and by how much?

A _____ B _____ b _____

6 Problem solving ★★★

A phone costs $1200. Three ways to pay:

1: cash, with a 5% discount
2: 24 payments of $55 per month
3: $200 deposit, then $45 per month for 24 months

a Find the total cost of each option.

b Find the difference between the cheapest and dearest options.

c Give one reason someone might still choose option 2.

1 _____ 2 _____ 3 _____ b _____

Comparing interest using technology

FAST FINISHERS

EXTENSION Finished exercise 1I? Try these. Show your working in each box.
In spreadsheet formulas * means × and ^ means "to the power of". Understanding Reasoning Problem solving

1 Understanding ★☆☆

This spreadsheet works out simple interest.

a What number will cell B4 show?

b Write a formula for B5 to give the total value.

	A	B
1	Principal	5000
2	Rate	0.04
3	Years	6
4	Interest	=B1*B2*B3
5	Total	

a _____ b _____

2 Understanding ★☆☆

Use the cells from question 1. Which formula gives the value of $5000 invested at 4% p.a. **compounded annually** for 6 years?

A `=B1*B2^B3`
B `=B1*(1+B2)^B3`
C `=B1*(1+B2*B3)`

a Choose the correct formula and find its value.

b What does formula C work out?

a _____ b _____

3 Reasoning ★★☆

Cell B3 contains =B2*1.05 and is filled down.

a What interest rate is used?

b Find the balances shown in B3 to B6.

	A	B
1	Year	Balance
2	0	2000
3	1	=B2*1.05
4	2	
5	3	
6	4	

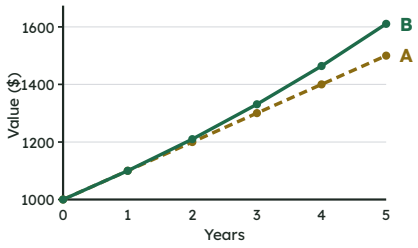
a _____

4 Problem solving ★★☆

$1000 is invested at 10% p.a. for 5 years. One graph is simple interest and one is compound interest.

a Which graph is compound interest? How can you tell?

b Find the difference in value after 5 years.



b _____

5 Reasoning Review 1G ★★☆

A spreadsheet finds compound interest paid **monthly** with the formula

`=B1*(1+B2/12)^(B3*12)`

B1 is the principal, B2 the yearly rate and B3 the number of years.

a Explain why it uses B2/12 and B3*12.

b Find the value for $3000 at 6% p.a. for 2 years.

b _____

6 Problem solving ★★★

Lucy has $8000 to invest.
Bank A: 4.5% p.a. simple interest
Bank B: 4.2% p.a. compounded annually

a Find the value at each bank after 4 years.

b Find the value at each bank after 5 years.

c Which bank is better for a long-term investment? Explain.

a _____ b _____

ANSWERS Worked answers. Check your method as well as your answer. Rounded answers may differ slightly in the last decimal place.

1A · Review of percentages

1. 3/8 = **0.375 = 37.5%**; 0.45 = **9/20 = 45%**; 2.5% = **0.025 = 1/40**; 1¼ = **1.25 = 125%**.
2. **a** 0.63, 0.625, 0.61, 0.667 (3 d.p.) **b** **61%, 5/8, 0.63, 2/3** **c** 7/20 = 35%, so **7/20 is larger**.
3. **a** 12% means 12/100, so multiply by 0.12 (or ÷ 100 × 12), not divide by 12. **b** 0.12 × 250 = **$30** **c** 10% = $25, 2% = $5, $25 + $5 = $30 ✓
4. **a** 9/28 × 100 = **32.1%** **b** 28 − 9 − 12 = 7; 7/28 = **25%** **c** Bus = 42.9%. 32.1 + 42.9 + 25 = 100% (all students counted once; rounding can make it 99.9% or 100.1%).
5. **a** A: 42/56 = **75%**; B: 33/42 = **78.6%** **b** **Test B**. The tests have different totals, so compare percentages, not raw marks.
6. **a** 36 ÷ 15 = **2.4** **b** 2.4 × 100 = **240** **c** 0.4 × 240 = **96** **d** 5% = 36 ÷ 3 = 12, and 40% = 8 × 5% = 96.

1B · Applying percentages

1. **a** 80 × 1.15 = **$92** **b** 64 × 0.75 = **$48** **c** **1.07** **d** **0.88**
2. **a** **8.5%** **b** 85/1000 = **17/200** **c** 3 ÷ 40 = 0.075 = **7.5%**
3. **a** A: 120 × 0.75 = **$90**; B: 108 − 20 = **$88** **b** **Shop B** is cheaper by **$2**, even though 25% sounds like more.
4. **a** 100% − 30% = **70%** **b** 630 ÷ 0.7 = **$900** **c** 630 × 1.3 = $819 ≠ $900. The 30% was taken from $900, not from $630.
5. **a** 48 × 1.1 = **$52.80** **b** 319 ÷ 1.1 = **$290** **c** 319 − 290 = **$29**
6. **a** 200 × 1.1 = 220; 220 × 0.9 = **$198** **b** The 10% decrease is taken from the bigger amount ($220), so $22 comes off but only $20 went on. **c** 198/200 = 0.99, so a **1% decrease**.

1C · Income

1. **a** 67 600 ÷ 52 = **$1300** **b** × 2 = **$2600** **c** 67 600 ÷ 12 = **$5633.33**
2. **a** 38 × 24.80 = **$942.40** **b** 1.5 × 24.80 = **$37.20** **c** 2 × 24.80 = **$49.60**
3. **a** 38 × 28 = **$1064** **b** 4 × 42 + 3 × 56 = 168 + 168 = **$336** **c** **$1400**
4. **a** 600 + 0.05 × 9000 = **$1050** **b** Needs 1150 − 600 = $550 commission; 550 ÷ 0.05 = **$11 000** **c** Any reasoned answer, e.g. Job A is safer; Job B is better if you sell more than $11 000 a week.
5. **a** 26 × 1.04 = **$27.04** **b** 35 × 27.04 = **$946.40** **c** 946.40 − 910 = **$36.40**
6. **a** 4 × 1.5 = 6, so it pays like 6 normal hours. **b** 36 + 6 = 42 hours; 1092 ÷ 42 = **$26** **c** 36 × 26 + 4 × 39 = 936 + 156 = $1092 ✓

1D · Income tax

1. **a** 58 000 − 1400 = **$56 600** **b** 4288 + 0.30 × 11 600 = 4288 + 3480 = **$7768**
2. **a** 0.16 × (30 000 − 18 200) = 0.16 × 11 800 = **$1888** **b** **$0** **c** $16 500 is under the $18 200 tax-free threshold.
3. **a** Only the dollars over $45 000 are taxed at 30c; the first $45 000 costs $4288. **b** 4288 + 0.3 × 5000 = **$5788**
4. **a** 4288 + 0.3 × 27 000 = **$12 388** **b** 0.02 × 72 000 = **$1440** **c** **$13 828** **d** 13 828 ÷ 72 000 = **19.2%**
5. **a** 26 × 420 = **$10 920** **b** 4288 + 0.3 × 17 000 = 9388; Medicare 1240; total **$10 628** **c** 10 920 − 10 628: **refund of $292**
6. **a** The tax on $45 000 is only $4288, and Tom paid more than that. **b** 6088 − 4288 = 1800; 1800 ÷ 0.30 = 6000; 45 000 + 6000 = **$51 000**

1E · Budgeting

1. **a** 45 × 52 = **$2340** **b** 1200 × 4 = **$4800** **c** 4800 ÷ 12 = **$400** **d** 180 ÷ 2 = **$90**
2. **a** 420 + 160 + 65 + 15 + 90 = **$750** **b** 1150 − 750 = **$400** **c** 420 ÷ 1150 = **36.5%**
3. **a** 30 × 23.50 = **$705** **b** 0.15 × 705 = **$105.75** **c** 1000 ÷ 105.75 = 9.46, so **10 weeks**
4. **a** 780 + 1152 + 900 + 2860 = **$5692** **b** 5692 ÷ 52 = **$109.46**
5. **a** 100 − 75 = **25%** **b** 0.1 × 1400 = **$140** **c** 2100 ÷ 140 = **15 fortnights**
6. **a** (1350 − 270) ÷ 12 = **$90** **b** She saves 310 − 235 = $75 a week. 270 + 12 × 75 = $1170, so **no** ($180 short). **c** 1080 ÷ 75 = 14.4, so **15 weeks**

ANSWERS Worked answers. Check your method as well as your answer. Rounded answers may differ slightly in the last decimal place.

1F · Simple interest

1. **a** 3000 × 0.04 × 5 = **$600** **b** **$3600**
2. **a** N = 1.5: 2500 × 0.06 × 1.5 = **$225** **b** N = 0.75: **$112.50** **c** Simple interest grows by the same amount each month, so half the time gives half the interest.
3. **a** The interest can't be 15 times the amount invested. **b** R must be a decimal: 0.05. 800 × 0.05 × 3 = **$120**
4. **a** 540 ÷ 3 = **$180** **b** 180 ÷ 4000 = 0.045 = **4.5% p.a.**
5. **a** 2000 × 0.035 = **$70** **b** 350 ÷ 70 = **5 years**
6. **a** 6000 × 0.08 × 3 = **$1440** **b** **$7440** **c** 7440 ÷ 36 = **$206.67** **d** **Yes**, $206.67 is less than $230.

1G · Compound interest

1. Year 1: $1000, $50, **$1050**. Year 2: $1050, $52.50, **$1102.50**. Year 3: $1102.50, $55.13, **$1157.63**
2. **a** 5000 × 1.06^4 = **$6312.38** **b** **$1312.38**
3. **a** 4000 × 0.05 × 3 = **$600** **b** 4000 × 1.05^3 = 4630.50, so **$630.50** **c** With compound interest you also earn interest on the interest from earlier years.
4. **a** 0.06 ÷ 12 = **0.005** **b** 2 × 12 = **24** **c** 2000 × 1.005^{24} = **$2254.32**
5. **a** 4% must be written as 0.04, so the bracket is 1.04, not 5. **b** 3000 × 1.04^2 = **$3244.80**
6. **a** 5000 × 1.08^5 = **$7346.64** **b** 9 years: $9995.02; 10 years: $10 794.62, so **10 years** **c** No. 18 years gives $19 980.10 and 19 years gives $21 578.60, so **19 years**. Once it reaches $10 000 it doubles again in about 9 more years, because each year's interest is bigger than the last.

1H · Investments and loans

1. **a** 380 × 48 = **$18 240** **b** 18 240 − 15 000 = **$3240**
2. **a** 0.1 × 1800 = **$180** **b** 180 + 85 × 24 = **$2220** **c** **$420**
3. **a** 3240 ÷ 4 = **$810** **b** 810 ÷ 15 000 = 0.054 = **5.4% p.a.**
4. Month 1: 1000, $10, 200, **$810.00**. Month 2: 810, $8.10, 200, **$618.10**. Month 3: 618.10, $6.18, 200, **$424.28**
5. **a** A: 10 000 × 0.042 × 3 = **$1260**; B: 10 000 × 1.04^3 − 10 000 = **$1248.64** **b** **Option A**, by **$11.36**
6. **a** 1: **$1140**; 2: **$1320**; 3: 200 + 1080 = **$1280** **b** 1320 − 1140 = **$180** **c** E.g. they don't have $1140 or $200 now, and $55 a month fits their budget.

1I · Comparing interest using technology

1. **a** 5000 × 0.04 × 6 = **1200** **b** **=B1+B4** (gives 6200)
2. **a** **B**: 5000 × 1.04^6 = **$6326.60** **b** C gives the simple interest total, $6200.
3. **a** **5% p.a.** compounded annually **b** **2100, 2205, 2315.25, 2431.01**
4. **a** **B.** It curves upward because the interest grows each year; A goes up $100 each year (straight line). **b** 1610.51 − 1500 = **$110.51**
5. **a** The yearly rate is split into 12 monthly rates, and there are 12 compounding periods each year. **b** 3000 × 1.005^{24} = **$3381.48**
6. **a** A: **$9440**; B: 8000 × 1.042^4 = **$9431.07** (A better) **b** A: **$9800**; B: **$9827.17** (B better) **c** **Bank B** for 5 years or more, because compound interest keeps growing faster.

ABOUT THE ARTWORK

The pictures in this booklet are parts of the Mandelbrot set, drawn by a computer that repeats one simple rule, $z \to z^2 + c$, hundreds of times for every point. This one is a spiral from the tail of a "seahorse" in Seahorse Valley. Zoom in on any part of the edge and you find more detail, forever. Maths can make beautiful things.

ACKNOWLEDGEMENT OF COUNTRY

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.